

图像处理：第五次作业

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1 Z 变换性质

(a) 若 $x(n)$ 的 Z 变换为 $X(z)$, 则 $(-1)^n x(n)$ 的 Z 变换为 $X(-z)$

由于 $x(n)$ 的 Z 变换为:

$$X(z) = \sum_{-\infty}^{\infty} x(n) z^{-n}$$

因为对称性质:

$$(-1)^n = (-1)^{-n}$$

所以 $(-1)^n x(n)$ 的 Z 变换为:

$$\sum_{-\infty}^{\infty} (-1)^n x(n) z^{-n} = \sum_{-\infty}^{\infty} (-1)^{-n} x(n) z^{-n} = \sum_{-\infty}^{\infty} x(n) (-z)^{-n} = X(-z)$$

(b) 若 $x(n)$ 的 Z 变换为 $X(z)$, 则 $x(-n)$ 的 Z 变换为 $X\left(\frac{1}{z}\right)$

由于:

$$X(z) = \sum_{-\infty}^{\infty} x(n) z^{-n}$$

令 $k = -n$, 则:

$$\sum_{-\infty}^{\infty} x(-n) z^{-n} = \sum_{-\infty}^{\infty} x(k) z^k$$

因此:

$$\sum_{-\infty}^{\infty} x(k) z^k = \sum_{-\infty}^{\infty} x(k) (z^{-1})^{-k} = X\left(\frac{1}{z}\right)$$

(c) 若 $x(n)$ 的 Z 变换为 $X(z)$, 则有信号“抽取”在 Z 域的性质

若 $x(n)$ 的 Z 变换为 $X(z)$, 则 $x_{\text{down}}(n) = x(2n) \Leftrightarrow X_{\text{down}}(z) = \frac{1}{2} [X(\sqrt{z}) + X(-\sqrt{z})]$

$$X_{\text{down}}(z) = \sum_{-\infty}^{\infty} x_{\text{down}}(n) z^{-n} = \sum_{-\infty}^{\infty} x(2n) z^{-n}$$

由于：

$$X(\sqrt{z}) = \sum_{-\infty}^{\infty} x(k)(\sqrt{z})^{-k} = \sum_{-\infty}^{\infty} x(k)z^{-k/2}$$

$$X(-\sqrt{z}) = \sum_{-\infty}^{\infty} x(k)(-\sqrt{z})^{-k} = \sum_{-\infty}^{\infty} x(k)(-1)^{-k}z^{-k/2} = \sum_{-\infty}^{\infty} x(k)(-1)^k z^{-k/2}$$

叠加后得到：

$$X(\sqrt{z}) + X(-\sqrt{z}) = \sum_{-\infty}^{\infty} x(k)z^{-k/2} + \sum_{-\infty}^{\infty} x(k)(-1)^k z^{-k/2}$$

$$= \sum_{-\infty}^{\infty} x(k) [1 + (-1)^k] z^{-k/2}$$

当 k 为偶数， $1 + (-1)^k = 2$ ；当 k 为奇数， $1 + (-1)^k = 0$ ，所以：

$$X(\sqrt{z}) + X(-\sqrt{z}) = \sum_{k \in \text{even}} x(k) \cdot 2 \cdot z^{-k/2}$$

令 $k = 2m$ ，其中 $m \in \mathbb{Z}$ ，则：

$$X(\sqrt{z}) + X(-\sqrt{z}) = 2 \cdot \sum_{-\infty}^{\infty} x(2m)z^{-m}$$

因此：

$$X_{\text{down}}(z) = \sum_{-\infty}^{\infty} x(2n)z^{-n} = \frac{1}{2} \left[2 \cdot \sum_{-\infty}^{\infty} x(2m)z^{-m} \right] = \frac{1}{2} [X(\sqrt{z}) + X(-\sqrt{z})]$$

2 正交镜像滤波器组推论

$$G_1(z) = -z^{-2k+1}G_0(-z^{-1}) \implies g_1(n) = (-1)^n g_0(2k-1-n)$$

由于：

$$G_1(z) = -z^{-2k+1}G_0(-z^{-1}) = -z^{-2k+1} \sum_{-\infty}^{\infty} g_0(n)(-z^{-1})^{-n} = \sum_{-\infty}^{\infty} g_0(n)(-1)^{n+1} z^{n+1-2k}$$

令 $-m = n + 1 - 2k$ ，则：

$$\sum_{-\infty}^{\infty} g_0(n)(-1)^{n+1} z^{n+1-2k} = \sum_{-\infty}^{\infty} g_0(2k-1-m)(-1)^{2k-m} z^{-m}$$

又因为：

$$(-1)^{2k-m} = (-1)^{2k} \cdot (-1)^{-m} = 1 \cdot (-1)^m = (-1)^m$$

所以：

$$\sum_{-\infty}^{\infty} g_0(2k-1-m)(-1)^{2k-m}z^{-m} = \sum_{-\infty}^{\infty} g_0(2k-1-m)(-1)^m z^{-m}$$

因此：

$$g_1(n) = (-1)^n g_0(2k-1-n)$$

3 正交族小波变换完美重建滤波器组推论

由于高通和低通综合滤波器交替反转关系：

$$g_1(n) = (-1)^n g_0(2k-1-n)$$

又因为分析滤波器是综合滤波器时域镜像正交关系：

$$\begin{cases} h_0(n) = g_0(2k-1-n) \\ h_1(n) = g_1(2k-1-n) \end{cases}$$

反转表达：

$$h_0(n) = g_0(2k-1-n) \implies h_0(2k-1-n) = g_0(n)$$

由：

$$h_1(n) = g_1(2k-1-n) = (-1)^{2k-1-n} g_0[2k-1-(2k-1-n)] = (-1)^{n+1} g_0(n)$$

即：

$$h_1(n) = (-1)^{n+1} h_0(2k-1-n)$$

4 哈尔小波变换矩阵

哈尔基函数：

$$h_k(z) = h_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2}, & \frac{q-1}{2^p} \leq z < \frac{q-0.5}{2^p} \\ -2^{p/2}, & \frac{q-0.5}{2^p} \leq z < \frac{q}{2^p} \\ 0, & z \in \left[0, \frac{q-1}{2^p}\right) \cup \left[\frac{q}{2^p}, 1\right] \end{cases}$$

根据公式 $k = 2^p + q - 1$ ，可得：

表 1: 哈尔变换参数组

k	p	q
0	0	0
1	0	1
2	1	1
3	1	2
4	2	1
5	2	2
6	2	3
7	2	4
8	3	1
9	3	2
10	3	3
11	3	4
12	3	5
13	3	6
14	3	7
15	3	8

以 $k = 6$ 为例，能满足条件的最大 2^p 是 4，所以 $p = 2$ ，因此 $q = k - 2^p + 1 = 3$ ，根据 $h_6(z)$ 哈尔基函数：

$$h_6(z) = \frac{1}{\sqrt{16}} \begin{cases} 2^{2/2}, & \frac{3-1}{2^2} \leq z < \frac{3-0.5}{2^2} \\ -2^{2/2}, & \frac{3-0.5}{2^2} \leq z < \frac{3}{2^2} \\ 0, & z \in \left[0, \frac{3-1}{2^2}\right) \cup \left[\frac{3}{2^2}, 1\right] \end{cases} = \begin{cases} 2, & 0.5 \leq z < 0.625 \\ -2, & 0.625 \leq z < 0.75 \\ 0, & z \in [0, 0.5) \cup [0.75, 1] \end{cases}$$

采样点 z 的取值即填充列，包含 $\frac{0}{16}, \frac{1}{16}, \frac{2}{16}, \dots, \frac{15}{16}$ 共 16 个值，因为 $\frac{8}{16} = 0.5, \frac{9}{16} =$

$0.5625, \frac{10}{16} = 0.625, \frac{11}{16} = 0.6875$, 所以哈尔变换矩阵第 6 行的 16 个元素为:

$$[0, 0, 0, 0, 0, 0, 0, 0, 2, 2, -2, -2, 0, 0, 0, 0]$$

以此类推, 可得 H_{16} 矩阵:

$$H_{16} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & 2 & -2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 \\ 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\sqrt{2} & -2\sqrt{2} \end{bmatrix}$$

5 二元组扩展与其系数

(a) $\varphi_0 = [1/\sqrt{2}, 1/\sqrt{2}]^\top, \varphi_1 = [1/\sqrt{2}, -1/\sqrt{2}]^\top$

$$\alpha_0 = \varphi_0^\top \cdot [3, 2]^\top = [1/\sqrt{2}, 1/\sqrt{2}] \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \frac{5\sqrt{2}}{2}$$

$$\alpha_1 = \varphi_1^\top \cdot [3, 2]^\top = [1/\sqrt{2}, -1/\sqrt{2}] \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \frac{\sqrt{2}}{2}$$

$$\alpha_0 \varphi_0 + \alpha_1 \varphi_1 = \frac{5\sqrt{2}}{2} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} + \frac{\sqrt{2}}{2} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = [3, 2]^\top$$

(b) $\varphi_0 = [1, 0]^\top, \varphi_1 = [1, 1]^\top$, 基础的对偶 $\bar{\varphi}_0 = [1, -1]^\top, \bar{\varphi}_1 = [0, 1]^\top$

$$\alpha_0 = \bar{\varphi}_0^\top \cdot [3, 2]^\top = [1, -1] \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 1$$

$$\alpha_1 = \bar{\varphi}_1^\top \cdot [3, 2]^\top = [0, 1] \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 2$$

$$\alpha_0\varphi_0 + \alpha_1\varphi_1 = 1 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = [3, 2]^\top$$

(c) $\varphi_0 = [1, 0]^\top$, $\varphi_1 = [-1/2, \sqrt{3}/2]^\top$, $\varphi_2 = [-1/2, -\sqrt{3}/2]^\top$, $\bar{\varphi}_i = 2\varphi_i/3$

$$\alpha_0 = \bar{\varphi}_0^\top \cdot [3, 2]^\top = \begin{bmatrix} 2/3 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 2$$

$$\alpha_1 = \bar{\varphi}_1^\top \cdot [3, 2]^\top = \begin{bmatrix} -1/3 \\ \sqrt{3}/3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = -1 + \frac{2\sqrt{3}}{3}$$

$$\alpha_2 = \bar{\varphi}_2^\top \cdot [3, 2]^\top = \begin{bmatrix} -1/3 \\ -\sqrt{3}/3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 2 \end{bmatrix} = -1 - \frac{2\sqrt{3}}{3}$$

$$\alpha_0\varphi_0 + \alpha_1\varphi_1 + \alpha_2\varphi_2 = 2 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \left(\frac{2\sqrt{3}}{3} - 1\right) \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix} - \left(\frac{2\sqrt{3}}{3} + 1\right) \begin{bmatrix} -1/2 \\ -\sqrt{3}/2 \end{bmatrix} = [3, 2]^\top$$

6 尺度函数未满足多分辨率分析的第二要求

$$\varphi(x) = \begin{cases} 1 & 0.25 \leq x < 0.75 \\ 0 & \text{other} \end{cases}$$

W_j 空间的所有 $k \in \mathbb{Z}$ 定义小波集合 $\{\psi_{j,k}(x)\}$:

$$\psi_{j,k}(x) = 2^{j/2}\psi(2^jx - k)$$

因此:

$$\varphi_{0,0}(x) = \varphi(x)$$

$$\varphi_{1,0}(x) = \sqrt{2}\varphi(2x)$$

$$\varphi_{1,1}(x) = \sqrt{2}\varphi(2x - 1)$$

因为 $\varphi_{0,0}(x)$ 的支撑集是 $[0.25, 0.75)$, $\varphi_{1,0}(x)$ 的支撑集是 $[0.125, 0.375)$, $\varphi_{1,1}(x)$ 的支撑集是 $[0.625, 0.875)$, 所以区间 $[0.375, 0.625)$ 出现空洞, 即 $\varphi_{1,0}(x) = \varphi_{1,1}(x) \equiv 0$, 而 $\varphi_{0,0}(x) \equiv 1$, 因此意味着无论这两个函数如何加权, 都无法填满中间这段空隙, 即 $\varphi_{0,0}(x)$ 无法由 $\varphi_{1,0}(x)$ 和 $\varphi_{1,1}(x)$ 加权求和表示。

7 离散小波变换

DWT 起始尺度 j 的函数:

$$W_{\varphi}(j_0, k) = \frac{1}{\sqrt{M}} \sum_x f(x) \varphi_{j_0, k}(x)$$

$$W_{\psi}(j, k) = \frac{1}{\sqrt{M}} \sum_x f(x) \psi_{j, k}(x)$$

$$f(0) = 1 \quad f(1) = 4 \quad f(2) = -3 \quad f(3) = 0$$

哈尔尺度函数:

$$\varphi(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 0 & \text{other} \end{cases}$$

哈尔小波函数:

$$\psi(x) = \begin{cases} 1 & 0 \leq x < 0.5 \\ -1 & 0.5 \leq x < 1 \\ 0 & \text{other} \end{cases}$$

小波集合:

$$\psi_{j, k}(x) = 2^{j/2} \psi \left(2^j \frac{x}{M} - k \right)$$

(a) 计算一维 DWT

若 $j_0 = 1$:

$$\begin{aligned} W_{\varphi}(1, 0) &= \frac{1}{\sqrt{4}} [f(0)\varphi_{1,0}(0) + f(1)\varphi_{1,0}(1) + f(2)\varphi_{1,0}(2) + f(3)\varphi_{1,0}(3)] \\ &= \frac{1}{2} (1 \times \sqrt{2} + 4 \times \sqrt{2} - 3 \times 0 + 0 \times 0) \\ &= \frac{5\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned}
W_{\varphi}(1, 1) &= \frac{1}{\sqrt{4}} [f(0)\varphi_{1,1}(0) + f(1)\varphi_{1,1}(1) + f(2)\varphi_{1,1}(2) + f(3)\varphi_{1,1}(3)] \\
&= \frac{1}{2} (1 \times 0 + 4 \times 0 - 3 \times \sqrt{2} + 0 \times \sqrt{2}) \\
&= -\frac{3\sqrt{2}}{2}
\end{aligned}$$

$$\begin{aligned}
W_{\psi}(1, 0) &= \frac{1}{\sqrt{4}} [f(0)\psi_{1,0}(0) + f(1)\psi_{1,0}(1) + f(2)\psi_{1,0}(2) + f(3)\psi_{1,0}(3)] \\
&= \frac{1}{2} [1 \times \sqrt{2} + 4 \times (-\sqrt{2}) - 3 \times 0 + 0 \times 0] \\
&= -\frac{3\sqrt{2}}{2}
\end{aligned}$$

$$\begin{aligned}
W_{\psi}(1, 1) &= \frac{1}{\sqrt{4}} [f(0)\psi_{1,1}(0) + f(1)\psi_{1,1}(1) + f(2)\psi_{1,1}(2) + f(3)\psi_{1,1}(3)] \\
&= \frac{1}{2} [1 \times 0 + 4 \times 0 - 3 \times \sqrt{2} + 0 \times (-\sqrt{2})] \\
&= -\frac{3\sqrt{2}}{2}
\end{aligned}$$

因此, DWT 集合是 $\left\{ \frac{5\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2} \right\}$ 。

(b) 根据变换值计算 $f(1)$

根据:

$$f(x) = \frac{1}{\sqrt{M}} \sum_k W_{\varphi}(j_0, k) \varphi_{j_0, k}(x) + \frac{1}{\sqrt{M}} \sum_{j=j_0}^{\infty} \sum_k W_{\psi}(j, k) \psi_{j, k}(x)$$

则有:

$$\begin{aligned}
f(x) &= \frac{1}{\sqrt{4}} [W_{\varphi}(1, 0)\varphi_{1,0}(x) + W_{\varphi}(1, 1)\varphi_{1,1}(x) + W_{\psi}(1, 0)\psi_{1,0}(x) + W_{\psi}(1, 1)\psi_{1,1}(x)] \\
&= \frac{1}{2} \left[\frac{5\sqrt{2}}{2}\varphi_{1,0}(x) - \frac{3\sqrt{2}}{2}\varphi_{1,1}(x) - \frac{3\sqrt{2}}{2}\psi_{1,0}(x) - \frac{3\sqrt{2}}{2}\psi_{1,1}(x) \right]
\end{aligned}$$

因此:

$$f(1) = \frac{1}{2} \times \left(\frac{5\sqrt{2}}{2} \times \sqrt{2} - \frac{3\sqrt{2}}{2} \times 0 - \frac{3\sqrt{2}}{2} \times \sqrt{2} - \frac{3\sqrt{2}}{2} \times 0 \right) = 1$$

8 完美重建来此小波变换分解后的滤波器输出

信号 $f = [1, 3, 5, 7, 4, 3, 2, 1]$ ，利用哈尔小波进行两层快速小波变换分解，计算各层的滤波器输出，然后再进行完美重建。其中：

$$\begin{aligned} \text{高通滤波器 } h_1 &= \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right] \\ \text{低通滤波器 } h_0 &= \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right] \end{aligned}$$

高通卷积：

$$f \circledast h_1 = \left[-\frac{1}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, \frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

低通卷积：

$$f \circledast h_0 = \left[\frac{1}{\sqrt{2}}, \frac{4}{\sqrt{2}}, \frac{8}{\sqrt{2}}, \frac{12}{\sqrt{2}}, \frac{11}{\sqrt{2}}, \frac{7}{\sqrt{2}}, \frac{5}{\sqrt{2}}, \frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

由于 $2 \downarrow$ 每隔一个点取一个值，所以：

$$\begin{aligned} W_\psi(1, n) &= \left\{ -\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\} \\ W_\varphi(1, n) &= \left\{ \frac{4}{\sqrt{2}}, \frac{12}{\sqrt{2}}, \frac{7}{\sqrt{2}}, \frac{3}{\sqrt{2}} \right\} \end{aligned}$$

上支路卷积：

$$W_\varphi(1, n) \circledast h_1 = \left[-2, -4, \frac{5}{2}, 2, \frac{3}{2} \right]$$

下支路卷积：

$$W_\psi(1, n) \circledast h_0 = \left[2, 8, \frac{19}{2}, 5, \frac{3}{2} \right]$$

由于 $2 \downarrow$ 每隔一个点取一个值，所以第二级：

$$\begin{aligned} W_\psi(0, n) &= \{-4, 2\} \\ W_\varphi(0, n) &= \{8, 5\} \end{aligned}$$

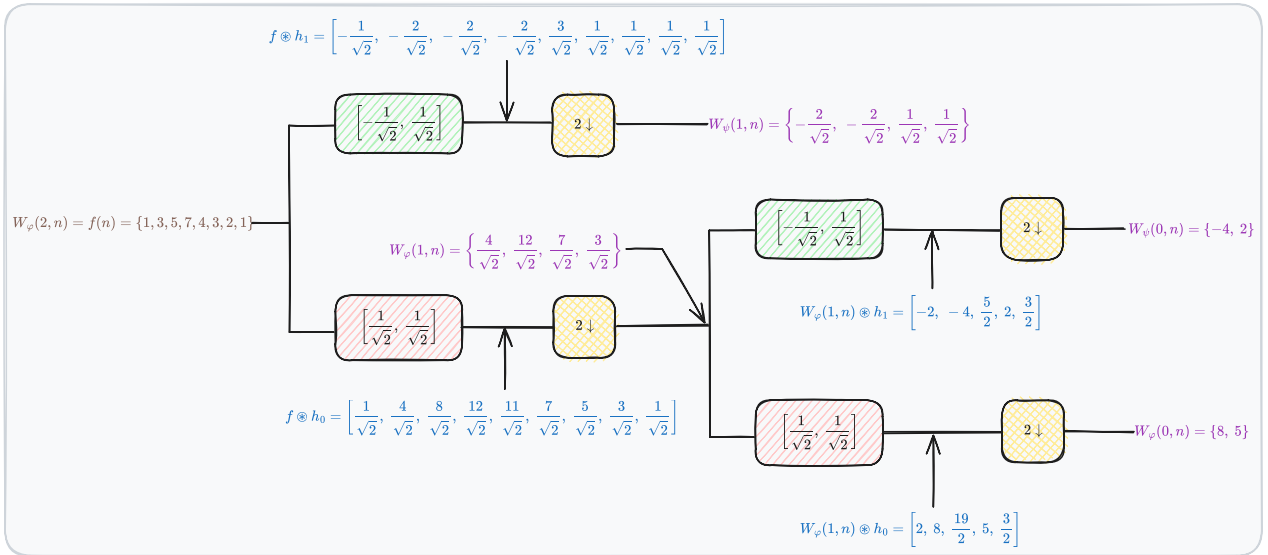


图 1: 用哈尔尺度和小波向量计算序列 f 的二尺度快速小波变换

$$\text{高通综合滤波器 } h_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{低通综合滤波器 } h_0 = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

由于 $2\uparrow$ 每隔一个点补充零值，所以：

$$W_\psi(0, n) = \{-4, 2\} \rightarrow \{-4, 0, 2, 0\}$$

$$W_\varphi(0, n) = \{8, 5\} \rightarrow \{8, 0, 5, 0\}$$

高通综合卷积：

$$\{-4, 0, 2, 0\} \otimes h_1 = \left[-\frac{4}{\sqrt{2}}, \frac{4}{\sqrt{2}}, \frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, 0 \right]$$

低通综合卷积：

$$\{8, 0, 5, 0\} \otimes h_0 = \left[\frac{8}{\sqrt{2}}, \frac{8}{\sqrt{2}}, \frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}}, 0 \right]$$

叠加：

$$W_\varphi(1, n) = \left[-\frac{4}{\sqrt{2}}, \frac{4}{\sqrt{2}}, \frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, 0 \right] + \left[\frac{8}{\sqrt{2}}, \frac{8}{\sqrt{2}}, \frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}}, 0 \right]$$

$$= \left[\frac{4}{\sqrt{2}}, \frac{12}{\sqrt{2}}, \frac{7}{\sqrt{2}}, \frac{3}{\sqrt{2}}, 0 \right]$$

由于 $2 \uparrow$ 每隔一个点补充零值，所以第二级：

$$\left[\frac{4}{\sqrt{2}}, \frac{12}{\sqrt{2}}, \frac{7}{\sqrt{2}}, \frac{3}{\sqrt{2}}, 0 \right] \rightarrow \left[\frac{4}{\sqrt{2}}, 0, \frac{12}{\sqrt{2}}, 0, \frac{7}{\sqrt{2}}, 0, \frac{3}{\sqrt{2}} \right]$$

第二级低通综合卷积：

$$\left[\frac{4}{\sqrt{2}}, 0, \frac{12}{\sqrt{2}}, 0, \frac{7}{\sqrt{2}}, 0, \frac{3}{\sqrt{2}} \right] \otimes h_0 = \left[2, 2, 6, 6, \frac{7}{2}, \frac{7}{2}, \frac{3}{2}, \frac{3}{2} \right]$$

又因为：

$$W_\psi(1, n) = \left\{ -\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right\}$$

所以 $2 \uparrow$ 每隔一个点补充零值，得到：

$$\left[-\frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right] \rightarrow \left[-\frac{2}{\sqrt{2}}, 0, -\frac{2}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right]$$

第二级高通综合卷积：

$$\left[-\frac{2}{\sqrt{2}}, 0, -\frac{2}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right] \otimes h_1 = \left[-1, 1, -1, 1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right]$$

第二级叠加：

$$\begin{aligned} W_\varphi(2, n) = f(n) &= \left[-1, 1, -1, 1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right] + \left[2, 2, 6, 6, \frac{7}{2}, \frac{7}{2}, \frac{3}{2}, \frac{3}{2} \right] \\ &= [1, 3, 5, 7, 4, 3, 2, 1] \end{aligned}$$

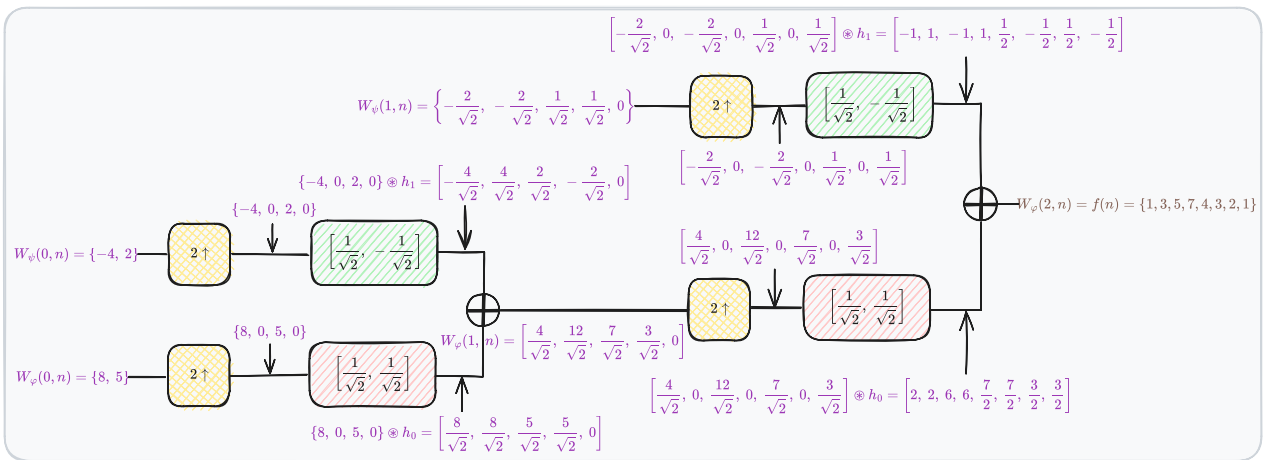


图 2: 用哈尔尺度和小波向量计算序列 $W_\psi(0, n), W_\varphi(0, n), W_\psi(1, n)$ 的二尺度快速小波逆变换